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Gas control systems for purging a printhead manufacturing apparatus

Abstract

A gas control system including a positive pressure vessel, a negative pressure vessel, a first pressure control valve configured to control a flow of gas to and from a first manifold of a printhead assembly, and a second pressure control valve configured to control a flow of gas to and from a second manifold of the printhead assembly. During a normal positive pressure mode, gas flows from the positive pressure vessel to the first manifold and the second manifold through a respective one of the first pressure control valve and the second pressure control valve. During a positive pressure purge mode, gas from the positive pressure vessel bypasses the first pressure control valve and the second pressure control valve to flow to the first manifold and the second manifold.

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Background/Summary

TECHNICAL FIELD

(1) The present specification generally relates to printing assemblies and, more specifically, to a control system of a printhead of a manufacturing apparatus and methods for using the same.

BACKGROUND

(2) Printing assemblies may be utilized to “build” an object from build material, such as three-dimensional objects or parts, in a layer-wise manner. Early iterations of printing assemblies were used for prototyping three-dimensional parts. However, as printing assembly technology has improved, there is an increased interest in utilizing printing assemblies for large-scale commercial production of parts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:
- (2) FIG. 1 schematically depicts a manufacturing apparatus including a gas control system and a printhead assembly, according to one or more embodiments shown and described herein;
- (3) FIG. 2 schematically depicts a manufacturing system for operating the manufacturing apparatus of FIG. 1, according to one or more embodiments shown and described herein;
- (4) FIG. 3 schematically depicts a perspective view of the gas control system of FIG. 1, according to one or more embodiments shown and described herein;
- (5) FIG. 4 schematically depicts components of the gas control system of FIG. 3 and a printhead of the printhead apparatus of FIG. 1, according to one or more embodiments shown and described herein;
- (6) FIG. 5 schematically depicts a flow of gas through the gas control system of FIG. 3 during a positive pressure purge mode, according to one or more embodiments shown and described herein;
- (7) FIG. 6 schematically depicts a flow of gas through the gas control system of FIG. 3 during a negative pressure purge mode, according to one or more embodiments shown and described herein;
- (8) FIG. 7 schematically depicts a flow of gas through the gas control system of FIG. 3 during a normal positive pressure mode, according to one or more embodiments shown and described herein;
- (9) FIG. 8 schematically depicts a flow of gas through the gas control system of FIG. 3 during a normal negative pressure mode, according to one or more embodiments shown and described herein;
- (10) FIG. 9 schematically depicts a flow of gas through the gas control system of FIG. 3 during an embodiment of a calibration mode, according to one or more embodiments shown and described herein;
- (11) FIG. 10 schematically depicts a flow of gas through the gas control system of FIG. 3 during another embodiment of a calibration mode, according to one or more embodiments shown and described herein; and
- (12) FIG. 11 schematically depicts a flow of gas through the gas control system of FIG. 3 during a cleaning mode, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(13) Generally, printing assemblies utilized in various contexts, including, for example, manufacturing applications, jet printing applications, and other printing types include parallel issues to those described above. For example, manufacturing apparatuses generally include printing assemblies, also referred to as printhead devices, which deposit materials through an array of jet nozzles during a manufacturing process. These printhead devices typically utilize a manifold having dual containers for inlet and outlet flow control to individual print heads. However, these nozzles may become clogged with binder such that a purging operation is required to alleviate the clog. Electric pump-based pressure control systems may be utilized to purge the printhead nozzles.

However, such electric pump-based systems are not well suited to work with binders in an inert gas environment. Specifically, electric pump-based systems cannot readily and safely be used with flammable materials, electric pump-based systems are generally slow to increase and decrease both pressure and flow. Additionally, electric pump-based systems are more prone to electro-mechanical component failure.

(14) Accordingly, a need exists for alternative printing assemblies and components thereof for controlling a pressure directed to a printhead so that clogs within a nozzle of the printhead may be addressed.

(15) Reference will now be made in detail to embodiments of manufacturing apparatuses, and components thereof, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. Embodiments of a manufacturing apparatus, an example of which is schematically depicted in FIG. 1, include a printhead assembly including a first manifold, a second manifold, and one or more printheads, a binder control system, a gas control system, a power source, and a binder tank. The power source provides power to the binder control system, the gas control system, and the printhead assembly. The binder control system controls the flow of binder from the binder tank to each of the first manifold and the second manifold. The gas control system controls a flow of gas from an external gas supply to each of the first manifold and the second manifold, as well as modulates the flow of gas during binder flow operations. During a purging operation, gas may be rapidly circulated between the gas control system and the manifolds to purge a clogged nozzle of the one or more printheads. Various embodiments of printing assemblies for manufacturing apparatuses, manufacturing apparatuses comprising the printing assemblies, and methods for using the same are described in further detail herein with specific reference to the appended drawings. It should be understood that the embodiments of the manufacturing apparatuses shown and described herein may be configured and operable to build three-dimensional and/or non-three dimensional objects or parts.

(16) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(17) It is noted that the terms “substantially” and “about” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

(18) Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation; and the number or type of embodiments described in the specification.

(19) As used herein, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a” component includes aspects

having two or more such components, unless the context clearly indicates otherwise.

(20) Embodiments described herein are directed to manufacturing apparatuses (e.g., additive manufacturing apparatuses) and components for manufacturing apparatuses, specifically printing assemblies for depositing binder and other jettable materials. Various embodiments described herein may be implemented to perform de-clogging operations of printing assemblies. It is understood that technology developed and described herein relates to manufacturing, however, aspects of the technology may have applications in related industries such as 2-D printing or the like.

(21) Referring now to FIG. 1, a manufacturing apparatus **100** is illustrated according to one or more embodiments described herein. The manufacturing apparatus **100** may generally include a printhead assembly **102**, a binder control system **104** for controlling a flow of binder to the printhead assembly **102**, a gas control system **106** for controlling a flow of gas to the printhead assembly **102**, a power source **108** for providing power to the binder control system **104**, the gas control system **106**, and the printhead assembly **102**, and a binder tank **110** for storing a supply of binder. The printhead assembly **102** includes a manifold assembly **112** and printhead(s) **114**. Although only one printhead **114** is illustrated in FIG. 1, it should be appreciated that the printhead assembly **102** may include any number of printheads including a plurality of printheads **114**.

(22) With respect to the manifold assembly **112**, the manifold assembly **112** includes an inlet or first manifold **116** and an outlet or second manifold **118**. In embodiments, the first manifold **116** stores binder prior to being delivered to the printhead(s) **114**. The second manifold **118** stores unused binder from the printhead(s) **114**. In some embodiments, the second manifold **118** receives binder directly from the first manifold **116** as opposed to unused binder that flowing through the printhead(s) **114**. In embodiments, the binder is a solvent-based binder.

(23) The first manifold **116** includes a first gas inlet line **120** placing the first manifold **116** in fluid communication with the gas control system **106**. The first gas inlet line **120** directs gas from the gas control system **106** to the first manifold **116** and includes a first gas gauge **122** for detecting a pressure of gas within the first manifold **116**.

(24) The first manifold **116** further also includes a first binder inlet line **124** and a first binder outlet line **126** placing the first manifold **116** in fluid communication with the binder control system **104**. The first binder inlet line **124** directs binder from the binder control system **104**, and thus the binder tank **110**, to the first manifold **116**. A first binder inlet valve **128** is provided in the first binder inlet line **124** for controlling a flow of binder to the first manifold **116**. The first binder outlet line **126** allows binder to flow out of the first manifold **116**. As such, binder is permitted to flow from an outlet **110A** of the binder tank **110**, through the binder control system **104**, and into the first manifold **116** through the first binder inlet line **124**. Similarly, binder is permitted to flow out of the first manifold **116** through the first binder outlet line **126**, back through the binder control system **104**, and return to an inlet **110B** of the binder tank **110**. A first binder outlet valve **130** is provided in the first binder outlet line **126** for controlling a flow of binder out of the first manifold **116**.

(25) The first manifold **116** further includes a first upper float **132A** and a first lower float **132B**. The first upper float **132A** and the first lower float **132B** are configured to detect a level of binder within the first manifold **116**. When the first upper float **132A** detects the level of binder exceeds a predetermined upper threshold, the gas control system **106** is operated, as discussed herein, to increase the pressure of gas within the first manifold **116**. Alternatively, when the first lower float **132B** detects the level of binder falls below a predetermined lower threshold, the gas control system **106** is operated to decrease the pressure of gas within the first manifold **116**.

(26) Similarly, the second manifold **118** includes a second gas inlet line **134** placing the second manifold **118** in fluid communication with the gas control system **106**. The second gas inlet line **134** directs gas from the gas control system **106** to the second manifold **118** and includes a second gas gauge **136** for detecting a pressure of gas within the second manifold **118**.

(27) The second manifold **118** further also includes a second binder inlet line **138** and a second binder outlet line **140** placing the second manifold **118** in fluid communication with the binder control system **104**. The second binder inlet line **138** directs binder from the binder control system **104**, and thus the binder tank **110**, to the second manifold **118**. A second binder inlet valve **142** is provided in the second binder inlet line **138** for controlling a flow of binder to the second manifold **118**. The second binder outlet line **140** allows binder to flow out of the second manifold **118**. As such, binder is permitted to flow from the outlet **110A** of the binder tank **110**, through the binder control system **104**, and into the second manifold **118** through the second binder inlet line **138**. Similarly, binder is permitted to flow out of the second manifold **118** through the second binder outlet line **140**, back through the binder control system **104**, and return to the inlet **110B** of the binder tank **110**. A second binder outlet valve **144** is provided in the second binder outlet line **140** for controlling a flow of binder out of the second manifold **118**. In embodiments, the second binder inlet line **138** merges with the first binder inlet line **124**, and the second binder outlet line **140** merges with the first binder outlet line **126**.

(28) The second manifold **118** further includes a second upper float **146A** and a second lower float **146B**. The second upper float **146A** and the second lower float **146B** are configured to detect a level of binder within the second manifold **118**. When the second upper float **146A** detects the level of binder exceeds a predetermined upper threshold, the gas control system **106** is operated, as discussed herein, to increase the pressure of gas within the second manifold **118**. Alternatively, when the second lower float **146B** detects the level of binder falls below a predetermined lower threshold, the gas control system **106** is operated to decrease the pressure of gas within the second manifold **118**. Although not shown, it should be appreciated that a bypass line may be provided for placing the first manifold **116** and the second manifold **118** in direct fluid communication with one another. The bypass line may permit binder to flow from the first manifold **116** to the second manifold **118** without having to flow through the printhead(s) **114**. This may be necessary to adjust a level of binder within either the first manifold **116** or the second manifold **118** without requiring additional binder to be pumped specifically to the first manifold **116** or the second manifold **118** from the binder tank **110**.

(29) The first manifold **116** is in fluid communication with the printhead(s) **114** via a first manifold supply line **148** and the second manifold **118** is in fluid communication with the printhead(s) **114** via a second manifold supply line **150**. A first manifold supply line valve **152** is provided in the first manifold supply line **148** for controlling a flow of binder between the printhead(s) **114** and the first manifold **116**. A second manifold supply line valve **154** is provided in the second manifold supply line **150** for controlling a flow of binder between the printhead(s) **114** and the second manifold **118**. In embodiments, binder flows from the first manifold **116** to the printhead(s) **114** via the first manifold supply line **148** and the binder is dispensed from one or more nozzles of the printhead(s) **114**. Thereafter, unused binder flows from the printhead(s) **114** to the second manifold **118** via the second manifold supply line **150**.

(30) The flow of binder through the printhead(s) **114** from the first manifold **116** and to the second manifold **118** is due in part to the pressure within the first manifold **116** being greater than the pressure within the second manifold **118**. For example, the pressure in the first manifold **116** may be between -150 mmH₂O and -250 mmH₂O, and the pressure in the second manifold **118** may be between -200 mmH₂O and -300 mmH₂O. More particular, for example, the pressure in the first manifold **116** may be between -175 mmH₂O and -225 mmH₂O, and the pressure in the second manifold **118** may be between -225 mmH₂O and -275 mmH₂O. However, the nozzles of the printhead(s) **114** may become clogged such that the printhead(s) **114** may need to undergo a purging operation to de-clog the nozzles. As described herein, the gas control system **106** may be operated to perform a purging operation such that the pressure within the first manifold **116** and the second manifold **118** may be repeatedly increased and decreased to de-clog the binder within the nozzles of the printhead(s) **114**. It should be appreciated that the pressures provided herein are

gauge pressures, i.e., relative to the environment surrounding the printhead assembly **102**.

(31) Referring now to FIG. 2, a manufacturing system **180** may be provided for operating the binder control system **104**, the gas control system **106**, the power source **108**, the binder tank **110**, and the printhead assembly **102** including the manifold assembly **112** and the printhead(s) **114**, as well as the individual components thereof, as discussed in more detail herein. The manufacturing system **180** may comprise an electronic control unit **182**, network interface hardware **184**, and a communication path **186** communicatively coupled these components.

(32) The electronic control unit **182** comprises a processor **188** and a non-transitory electronic memory **190** to which various components are communicatively coupled. In some embodiments, the processor **188** and the non-transitory electronic memory **190** and/or the other components are included within a single device. In other embodiments, the processor **188** and the non-transitory electronic memory **190** and/or the other components may be distributed among multiple devices that are communicatively coupled. The electronic control unit **182** includes non-transitory electronic memory **190** that stores a set of machine-readable instructions. The processor **188** executes the machine-readable instructions stored in the non-transitory electronic memory **190**. The non-transitory electronic memory **190** may comprise RAM, ROM, flash memories, hard drives, or any device capable of storing machine-readable instructions such that the machine-readable instructions can be accessed by the processor **188**. Accordingly, the manufacturing system **180** described herein may be implemented in any conventional computer programming language, as pre-programmed hardware elements, or as a combination of hardware and software components. The non-transitory electronic memory **190** may be implemented as one memory module or a plurality of memory modules.

(33) In some embodiments, the non-transitory electronic memory **190** includes instructions for executing the functions of the manufacturing system **180**. The instructions may include instructions for operating the binder control system **104**, the gas control system **106**, the power source **108**, the binder tank **110**, and the printhead assembly **102** including the manifold assembly **112** and the printhead(s) **114**, as well as the individual components thereof.

(34) The processor **188** may be any device capable of executing machine-readable instructions. For example, the processor **188** may be an integrated circuit, a microchip, a computer, or any other computing device. The non-transitory electronic memory **190** and the processor **188** are coupled to the communication path **186** that provides signal interconnectivity between various components and/or modules of the manufacturing system **180**. Accordingly, the communication path **186** may communicatively couple any number of processors with one another, and allow the modules coupled to the communication path **186** to operate in a distributed computing environment. Specifically, each of the modules may operate as a node that may send and/or receive data. As used herein, the term “communicatively coupled” means that coupled components are capable of exchanging data signals with one another such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides, and the like.

(35) As schematically depicted in FIG. 2, the communication path **186** communicatively couples the processor **188** and the non-transitory electronic memory **190** of the electronic control unit **182** with a plurality of other components of the manufacturing system **180**. For example, the manufacturing system **180** depicted in FIG. 2 includes the processor **188** and the non-transitory electronic memory **190** communicatively coupled with the binder control system **104**, the gas control system **106**, the power source **108**, the binder tank **110**, the manifold assembly **112**, and the printhead(s) **114**.

(36) In some embodiments, the manufacturing system **180** includes network interface hardware **184** for communicatively coupling the manufacturing system **180** to a computing device via a network. The computing device may include, without limitation, a smartphone, a tablet, a personal media player, or any other electric device that includes wireless communication functionality. It is to be appreciated that, when provided, the computing device may serve to provide user commands to the

electronic control unit **182** remotely by wirelessly communicating with the electronic control unit **182** via a network. In embodiments, the manufacturing system **180** may include an operating device allowing for a user to control operation of the binder control system **104**, the gas control system **106**, the power source **108**, the binder tank **110**, the manifold assembly **112**, and the printhead(s) **114**. In some embodiments, the operating device may be a switch, toggle, button, or any combination of controls to provide user operation. The operating device is coupled to the communication path **186** such that the communication path **186** communicatively couples the operating device to other modules of the manufacturing system **180**.

(37) Referring now to FIGS. **3** and **4**, the gas control system **106** includes a panel **200** and a plurality of various components mounted to the panel **200** in any suitable manner such as by utilizing brackets and other fasteners. The components generally include a positive pressure vessel **202** for providing a gas, such as an inert gas, to each of the first manifold **116** and second manifold **118** of the manifold assembly **112**, a negative pressure vessel **204** for drawing a gas, such as an inert gas, out of each of the first manifold **116** and the second manifold **118** of the manifold assembly **112**, a first pressure control valve **206** for controlling the pressure within the first manifold **116**, and a second pressure control valve **208** for controlling the pressure within the second manifold **118**. It will be understood that the positive pressure vessel **202** may be any suitable vessel configured to define a plenum chamber containing a gas at positive pressure. It will be understood that the negative pressure vessel **204** may be any suitable vessel configured to define a vacuum chamber containing a gas at negative pressure.

(38) The first pressure control valve **206** and the second pressure control valve **208** control the amount of gas allowed to flow into and out of, or to and from, the first manifold **116** and the second manifold **118**, respectively, by controlling the rate of gas flowing through the first pressure control valve **206** and the second pressure control valve **208**. In embodiments, the first pressure control valve **206** and the second pressure control valve **208** are proportional valves. Prior to performing a purging operation of the manifold assembly **112** or during normal operation of the gas control system **106**, the positive pressure vessel **202** is positively pressurized, e.g., above 0 psi, and the negative pressure vessel **204** is negatively pressurized to provide a vacuum, i.e., below 0 psi. More particularly, for example, the positive pressure vessel **202** may be positively pressurized to a pressure greater than or equal to 10 psi and less than or equal to 20 psi, and the negative pressure vessel **204** may be negatively pressurized to a pressure greater than or equal to -10 psi and less than or equal to -15 psi relative to atmosphere. However, after a purging operation when the positive pressure vessel **202** releases gas to the first manifold **116** and the second manifold **118**, and the negative pressure vessel **204** recovers the gas, the positive pressure vessel **202** and the negative pressure vessel **204** may become depressurized, i.e., the pressure within the positive pressure vessel **202** and the negative pressure vessel **204** is 0 psi. Thus, as discussed in more detail herein, the positive pressure vessel **202** and the negative pressure vessel **204** must be recalibrated from a depressurized state to a calibrated state to perform a subsequent purging operation or continue performing a normal operation.

(39) In embodiments, the gas control system **106** further includes a first positive pressure valve **210**, a control side first manifold valve **212**, a first negative pressure valve **214**, and a control side second manifold valve **216**. The first positive pressure valve **210** is positionable between a first open state, a second open state, and a closed state. When the first positive pressure valve **210** is in the closed state, gas may be prevented from flowing through the first positive pressure valve **210** from an outlet **202B** of the positive pressure vessel **202**.

(40) When the first positive pressure valve **210** is in the first open state, gas from the positive pressure vessel **202** is permitted to flow through the first positive pressure valve **210** and distributed to both the control side first manifold valve **212** and the control side second manifold valve **216** via a first positive pressure distributor **218**. The first positive pressure distributor **218** receives gas from the positive pressure vessel **202** via the first positive pressure valve **210** and

redirects the gas to the control side first manifold valve **212** and the control side second manifold valve **216**. In embodiments, the first positive pressure distributor **218** may include a controllable valve feature to control the amount of gas flowing to either one or both of the control side first manifold valve **212** and the control side second manifold valve **216**. In other embodiments, the first positive pressure distributor **218** may be fixed such that gas is evenly distributed to the control side first manifold valve **212** and the control side second manifold valve **216**.

(41) When the first positive pressure valve **210** is in the second open state, the gas from the positive pressure vessel **202** is permitted to flow through the first positive pressure valve **210** and distributed to both the first pressure control valve **206** and the second pressure control valve **208** via a second positive pressure distributor **220**. The second positive pressure distributor **220** receives gas from the positive pressure vessel **202** via the first positive pressure valve **210** and redirects the gas to the first pressure control valve **206** and the second pressure control valve **208**. Similar to the first positive pressure distributor **218**, in embodiments, the second positive pressure distributor **220** may include a controllable valve feature to control the amount of gas to either one or both of the first pressure control valve **206** and the second pressure control valve **208**. In other embodiments, the second positive pressure distributor **220** may be fixed such that gas is evenly distributed to the first pressure control valve **206** and the second pressure control valve **208**. As discussed in more detail herein, the first pressure control valve **206** and the second pressure control valve **208** control the amount of gas being directed to the first manifold **116** and the second manifold **118** from the positive pressure vessel **202**.

(42) The control side first manifold valve **212** is positionable between a first open state, a second open state, and a closed state. When the control side first manifold valve **212** is in the closed state, the gas, either provided to the first manifold **116** by the positive pressure vessel **202** or drawn from the first manifold **116** by the negative pressure vessel **204**, is prohibited from flowing through the control side first manifold valve **212**. When the control side first manifold valve **212** is in the first open state, the gas, either provided to the first manifold **116** by the positive pressure vessel **202** or drawn from the first manifold **116** by the negative pressure vessel **204**, is permitted to flow through the control side first manifold valve **212**. More particularly, when the control side first manifold valve **212** is in the first open state, gas provided by the first positive pressure valve **210** is permitted to flow through the control side first manifold valve **212** from the first positive pressure valve **210** to a printhead side first manifold valve **222**. When the control side first manifold valve **212** is in the second open state, gas is directed through the control side first manifold valve **212** and to the first pressure control valve **206**. However, when the printhead side first manifold valve **222** is positioned in a closed state, gas is prohibited from flowing through the printhead side first manifold valve **222** to or from the first manifold **116**.

(43) Similarly, the control side second manifold valve **216** is positionable between a first open state, a second open state, and a closed state. When the control side second manifold valve **216** is in the closed state, the gas, either provided to the second manifold **118** by the positive pressure vessel **202** or drawn from the second manifold **118** by the negative pressure vessel **204**, is prohibited from flowing through the control side second manifold valve **216**. When the control side second manifold valve **216** is in the first open state, the gas, either provided to the second manifold **118** by the positive pressure vessel **202** or drawn from the second manifold **118** by the negative pressure vessel **204**, is permitted to flow through the control side second manifold valve **216**. More particularly, when the control side second manifold valve **216** is in the first open state, gas provided by the first positive pressure valve **210** is permitted to flow through the control side second manifold valve **216** from the first positive pressure valve **210** to a printhead side second manifold valve **224**. When the control side second manifold valve **216** is in the second open state, gas is directed through the control side second manifold valve **216** and to the second pressure control valve **208**. However, when the printhead side second manifold valve **224** is positioned in a closed state, gas is prohibited from flowing through the printhead side second manifold valve **224** to or

from the second manifold **118**.

(44) The first negative pressure valve **214** is also positionable between an open state and a closed state. When the first negative pressure valve **214** is in the closed state, gas may be prevented from flowing through the first negative pressure valve **214** to an inlet **204A** of the negative pressure vessel **204**.

(45) When the first negative pressure valve **214** is in the open state, gas, i.e., a secondary flow of gas, is permitted to flow through the first negative pressure valve **214**, such as from a negative pressure distributor **226**. The negative pressure distributor **226** receives gas flowing from each of the first pressure control valve **206** and the second pressure control valve **208**, which is received from the first manifold **116** and the second manifold **118**, respectively. As discussed in more detail herein, the first pressure control valve **206** and the second pressure control valve **208** control the amount of gas being drawn out of the first manifold **116** and the second manifold **118**, and directed to the negative pressure vessel **204** during a vacuum recovery mode or a normal operating mode.

(46) In embodiments, the gas control system **106** includes a positive pressure meter **228** through which gas flows from the positive pressure vessel **202** to the first positive pressure valve **210**. A positive pressure sensor **230** is in fluid communication with the positive pressure meter **228** to detect a rate of flow of gas through the positive pressure meter **228**. In embodiments, during a power failure or complete system shutdown, the positive pressure meter **228** may be operated to direct all gas from the positive pressure vessel **202** to an external gas receptacle **232** so that the gas control system **106** may be depressurized. An external valve **234** may be provided for permitting the flow of gas from the positive pressure meter **228** to the external gas receptacle **232**.

(47) Similarly, in embodiments, the gas control system **106** also includes a negative pressure meter **236** through which gas flows from first negative pressure valve **214** to the negative pressure vessel **204**. A negative pressure sensor **238** is in fluid communication with the negative pressure meter **236** to detect a rate of flow of gas through the negative pressure meter **236**.

(48) The components described above control the flow of gas between each of the positive pressure vessel **202** and the negative pressure vessel **204** and the manifolds. However, as discussed herein, the positive pressure vessel **202** and the negative pressure vessel **204** will need to be recalibrated after a purging operation from a depressurized state to a calibrated state such that the positive pressure vessel **202** contains a positive pressure of gas, above 0 psi, and the negative pressure vessel **204** is depressurized to have a negative pressure/vacuum, i.e., below 0 psi. As such, the following components described herein are utilized for calibrating both the positive pressure vessel **202** and the negative pressure vessel **204** from the depressurized state to the calibrated state.

(49) In embodiments, the gas control system **106** further includes a second positive pressure valve **240** for permitting gas to flow into the positive pressure vessel **202**, such as during a cleaning operation or when calibrating the positive pressure vessel **202** back into the calibrated state. The second positive pressure valve **240** is positionable between an open state and a closed state. When the second positive pressure valve **240** is in the open state, gas is permitted to flow into the inlet **202A** of the positive pressure vessel **202**. When the second positive pressure valve **240** is in the closed state, gas is prohibited from flowing into the inlet **202A** of the positive pressure vessel **202**. Similarly, the gas control system **106** further includes a second negative pressure valve **242** for permitting gas to flow both into an outlet **204B** of the negative pressure vessel **204**, such as during a cleaning operation, and out of the outlet **204B** of the negative pressure vessel **204**, such as to create a vacuum within the negative pressure vessel **204** and calibrate the negative pressure vessel **204** back into the calibrated state. The second negative pressure valve **242** is positionable between an open state and a closed state. When the second negative pressure valve **242** is in the open state, gas is permitted to flow through the second negative pressure valve **242**, such as to or from the outlet **204B** of the negative pressure vessel **204**. When the second negative pressure valve **242** is in the closed state, gas is prohibited from flowing to or from the outlet **204B** of the negative pressure vessel **204**.

(50) After a purging operation, discussed in more detail herein, when the positive pressure vessel **202** and the negative pressure vessel **204** are both in the depressurized state, gas may be redistributed from the negative pressure vessel **204** to the positive pressure vessel **202** to calibrate the positive pressure vessel **202** and the negative pressure vessel **204** into respective calibrated states such that the positive pressure vessel **202** is positively pressurized and the negative pressure vessel **204** is negatively pressurized. In embodiments, as discussed in more detail herein, the gas control system **106** may include a dual piston assembly **244**, a compressed air tank **246**, a first piston valve **248**, and a second piston valve **250**. The dual piston assembly **244** is operated to draw gas out of the outlet **204B** of the negative pressure vessel **204** and direct the gas into the inlet **202A** of the positive pressure vessel **202**. More particularly, in embodiments, the dual piston assembly **244** includes a piston and operates by repeatedly alternating the piston between a first position and a second position to create a suction force drawing the gas from the compressed air tank **246** and directing the gas toward the positive pressure vessel **202**. Although the dual piston assembly **244** is shown in the same position in FIGS. **6-11**, it should be appreciated that when the dual piston assembly **244** is operated, the piston is repeatedly actuated. The dual piston assembly **244** may be operated to circulate gas between the positive pressure vessel **202** and the negative pressure vessel **204** to achieve a fully closed gas circuit. This reduces the degree of risk to an operator.

(51) The first piston valve **248** may be positionable between an open state and a closed state. When the first piston valve **248** is in the open state, compressed air from the compressed air tank **246** may be delivered to the dual piston assembly **244** to actuate the dual piston assembly **244**. When the first piston valve **248** is in the closed state, compressed air may not be delivered to the dual piston assembly **244**. When the second piston valve **250** is in the open state, gas is permitted to both enter the dual piston assembly **244** from the negative pressure vessel **204**, so as to enter the dual piston assembly **244**, and exit the dual piston assembly **244**, so as to be delivered to the positive pressure vessel **202**. When the second piston valve **250** is in the closed position, gas is prohibited from entering the dual piston assembly **244** from the negative pressure vessel **204** and gas is prohibited from being delivered to the positive pressure vessel **202** from the dual piston assembly **244**.

(52) In another embodiment, the gas control system **106** includes an external gas supply **252** for delivering gas to the positive pressure vessel **202** independent of gas being drawn out of the negative pressure vessel **204**. The external gas supply **252** may be configured to deliver gas to both of the positive pressure vessel **202** and the negative pressure vessel **204**, such as during the cleaning operation. In this embodiment, and discussed in more detail herein, the external gas supply **252** provides gas to the positive pressure vessel **202** and the negative pressure vessel **204** via an external gas supply distributor **254**. The external gas supply distributor **254** receives gas from the external gas supply **252** and redirects the gas to both the positive pressure vessel **202** and the negative pressure vessel **204**. In embodiments, the external gas supply distributor **254** may include a controllable valve feature to control the amount of gas flowing to either one or both of the positive pressure vessel **202** and the negative pressure vessel **204**. In other embodiments, the external gas supply distributor **254** may be fixed such that gas is evenly distributed to the positive pressure vessel **202** and the negative pressure vessel **204**.

(53) In other embodiments, the gas control system **106** may include a Venturi device **256** configured to receive gas or other fluid from the external gas supply **252** and draw gas out of the negative pressure vessel **204** to create a negative pressure within the negative pressure vessel **204**, thereby placing the negative pressure vessel **204** back into the calibrated state. The gas control system **106** may include a first Venturi valve **258** provided between the external gas supply **252** and the Venturi device **256**, and positionable between an open and a closed position. When the first Venturi valve **258** is in the open position, fluid flow from the external gas supply **252** is permitted to flow into the Venturi device **256**. When the first Venturi valve **258** is in the closed position, fluid flow from the external gas supply **252** is prohibited from entering the Venturi device **256**. In embodiments, the gas control system **106** further includes a second Venturi valve **260** provided

between the Venturi device **256** and the negative pressure vessel **204**. The second Venturi valve **260** is configured to allow gas to flow in only one direction between the Venturi device **256** and the negative pressure vessel **204**. Specifically, the second Venturi valve **260** permits gas to be drawn out of the negative pressure vessel **204** and provided to the Venturi device **256**, but not flow back into the negative pressure vessel **204**.

(54) The gas control system **106** is operable between multiple modes such as, for example, a positive pressure purge mode during a purging operation, a negative pressure purge mode or vacuum recovery mode during a purging operation, a normal positive pressure mode during a normal operation, a normal negative pressure mode during a normal operation, a calibration mode, and a cleaning mode.

(55) As described herein, the manufacturing system **180** includes an electronic control unit **182**. It should be appreciated that the electronic control unit **182** is configured to operate the individual components of the gas control system **106**. For example, the electronic control unit **182** is responsible for controlling at least the first positive pressure valve **210**, the control side first manifold valve **212**, the first negative pressure valve **214**, the control side second manifold valve **216**, the first pressure control valve **206**, and the second pressure control valve **208** between the various operating positions and states.

(56) Referring now to FIG. 5, a flow of gas during the positive pressure purge mode of a purging operation is depicted in solid line. Specifically, when it is determined that the printhead(s) **114** of the printhead assembly **102** require purging, such as when a clog is detected within one or more nozzles of printhead(s) **114**, the gas control system **106** performs the purging operation. The purging operation includes a positive pressure purge mode and a negative pressure purge mode, also referred to herein as a vacuum recovery mode. During the positive pressure purge mode of the purging operation, gas from the positive pressure vessel **202** is released into the first manifold **116** and the second manifold **118**. More particularly, during the positive pressure purge mode, the first positive pressure valve **210** is positioned into the first open state in which gas is directed to both of the control side first manifold valve **212** and the control side second manifold valve **216** through the first positive pressure distributor **218**. In the positive pressure purge mode, the control side first manifold valve **212** and the control side second manifold valve **216** are both positioned into the first open state such that gas is permitted to flow to the printhead side first manifold valve **222** and the printhead side second manifold valve **224**. Thereafter, the printhead side first manifold valve **222** and the printhead side second manifold valve **224** are both positioned into the open state to permit the gas to flow into the first manifold **116** and the second manifold **118**, respectively. By allowing gas to flow, unrestricted, from the positive pressure vessel **202** to both the first manifold **116** and the second manifold **118**, the pressure within the first manifold **116** and the second manifold **118** is increased to push additional binder through to the printhead(s) **114**. Additionally, the gas is permitted to bypass the first pressure control valve **206** and the second pressure control valve **208**.

(57) Referring now to FIG. 6, a flow of gas during the negative pressure purge mode or vacuum recovery mode of the purging operation is depicted in solid line. After the positive pressure purge mode is completed, such that the positive pressure vessel **202** is depressurized and the gas is delivered to the first manifold **116** and the second manifold **118**, the gas control system **106** is operated to perform the negative pressure purge mode, or vacuum recovery mode. As discussed above, during the positive pressure purge mode, the printhead side first manifold valve **222**, the printhead side second manifold valve **224**, the control side first manifold valve **212**, and the control side second manifold valve **216** are each in the open state. However, in the negative pressure purge mode, the control side first manifold valve **212** and the control side second manifold valve **216** are positioned into the second open state such that gas from the first manifold **116** and the second manifold **118** may be directed to the first pressure control valve **206** and the second pressure control valve **208**, respectively. The first negative pressure valve **214** is then positioned into the

open state such that the vacuum within the negative pressure vessel **204** draws gas out of each of the first manifold **116** and the second manifold **118** and through the first pressure control valve **206** and the second pressure control valve **208**. During the negative pressure purge mode, the first pressure control valve **206** and the second pressure control valve **208** may be configured to permit full, unrestricted flow of gas from the first manifold **116** and the second manifold **118** to the negative pressure vessel **204**. It should be appreciated that gas does not flow through the first pressure control valve **206** and the second pressure control valve **208** when in the positive pressure purge mode to provide a direct flow of gas to the first manifold **116** and the second manifold **118**, and further eliminate the need for additional gas lines connected to the first pressure control valve **206** and the second pressure control valve **208**.

(58) It should be appreciated that, during the purging operation, the gas control system **106** may repeatedly alternate between the positive pressure purging state and the negative pressure purging state a predetermined number of times within short intervals of one another to purge the printhead(s) **114**.

(59) Referring now to FIG. 7, a flow of gas during the normal positive pressure mode of a normal operation is depicted. During normal operation of the gas control system **106**, gas is provided to the first manifold **116** and the second manifold **118** to maintain a level of binder within the first manifold **116** and the second manifold **118** within a predetermined range. This differs from the positive pressure purge mode in which an increased amount of gas is provided to the first manifold **116** and the second manifold **118** to clear a clog. As noted hereinabove, the level of binder within the first manifold **116** is determined by a first upper float **132A** and a first lower float **132B**. Similarly, the level of binder within the second manifold **118** is determined by a second upper float **146A** and a second lower float **146B**. When the level of binder within either the first manifold **116** or the second manifold **118** rises above the predetermined range, the pressure within the corresponding first manifold **116** or the second manifold **118** is increased. For example, if the level of binder within both the first manifold **116** and the second manifold **118** rises above the predetermined range, the pressure within both the first manifold **116** and the second manifold **118** is increased by directing gas from the positive pressure vessel **202** to the first manifold **116** and the second manifold **118**. More particularly, in the present example, the first positive pressure valve **210** is positioned into the second open state to allow gas to flow from the positive pressure vessel **202**. Thereafter, the gas, i.e., a primary flow of gas, which diverges to form a first flow of gas and a second flow of gas, is directed to both the first pressure control valve **206** and the second pressure control valve **208** through the first positive pressure distributor **218**. The first pressure control valve **206** and the second pressure control valve **208** are then operated to control the amount of gas permitted to flow to the first manifold **116** and the second manifold **118**, respectively, based on the level of binder detected within each of the first manifold **116** and the second manifold **118**. For example, if the level of binder is higher in the first manifold **116** as compared to the level of binder in the second manifold **118**, the first pressure control valve **206** may permit a greater amount of gas to flow to the first manifold **116** as compared to the amount of gas permitted to flow to the second manifold **118** by the second pressure control valve **208**. During the normal positive pressure mode, the control side first manifold valve **212** and the control side second manifold valve **216** are both positioned into the second open state to permit gas to flow from the first pressure control valve **206** and the second pressure control valve **208**. Additionally, the printhead side first manifold valve **222** and the printhead side second manifold valve **224** are both positioned into the open state to permit the gas to enter the first manifold **116** and the second manifold **118**, respectively.

(60) Referring now to FIG. 8, a flow of gas during the normal negative pressure mode of the normal operation is depicted. It should be appreciated that the direction of flow of gas during the normal negative pressure mode of the normal operation is the same as the direction of flow of gas during the negative pressure purge mode of the purging operation. During normal operation of the gas control system **106** while in the normal negative pressure mode, gas is drawn the first manifold

116 and the second manifold **118** to maintain the level of binder within the first manifold **116** and the second manifold **118** within the predetermined range. This differs from the negative pressure purge mode in which an increased amount of gas is drawn from the first manifold **116** and the second manifold **118** to clear a clog. In instances in which the level of binder detected in the first manifold **116** or the second manifold **118** is determined to fall below the predetermined range, the pressure in the corresponding first manifold **116** or second manifold **118** must be reduced to permit the level of binder to increase. For example, if the level of binder within both the first manifold **116** and the second manifold **118** falls below the predetermined range, the pressure within both the first manifold **116** and the second manifold **118** is decreased by drawing gas out of the first manifold **116** and the second manifold **118** by the negative pressure vessel **204**. During the normal negative pressure mode, the control side first manifold valve **212** and the control side second manifold valve **216** are both positioned in the second open state to permit gas to flow from to the first pressure control valve **206** and the second pressure control valve **208**. Additionally, the printhead side first manifold valve **222** and the printhead side second manifold valve **224** are both positioned into the open state to permit the gas to be drawn from the first manifold **116** and the second manifold **118**, respectively. More particularly, in the present example, the first negative pressure valve **214** is positioned into the open state to allow gas to flow into the negative pressure vessel **204** caused by the negative pressure or vacuum created within the negative pressure vessel **204**. Thereafter, the first pressure control valve **206** and the second pressure control valve **208** are configured to direct the flow of gas from the first manifold **116** and the second manifold **118** to the negative pressure vessel **204** through the negative pressure distributor **226**. More particularly, the first pressure control valve **206** and the second pressure control valve **208** are operated to control the amount of gas permitted to be drawn from the first manifold **116** and the second manifold **118** based on the level of binder detected within each of the first manifold **116** and the second manifold **118**. For example, if the level of binder is higher in the first manifold **116** as compared to the level of binder in the second manifold **118**, the first pressure control valve **206** may permit less gas to be drawn out of the first manifold **116** as compared to the amount of gas permitted to be drawn out of the second manifold **118** by the second pressure control valve **208**.

(61) It should be appreciated that the gas control system **106** may alternate between the normal positive pressure mode and the normal negative pressure mode at any point to adjust the pressure within the first manifold **116** and the second manifold **118**. In addition, it should be appreciated that, in embodiments, the gas control system **106** may be operated to allow for gas to flow into one of the first manifold **116** or the second manifold **118** from the positive pressure vessel **202** while gas is being drawn from the other of the first manifold **116** or the second manifold **118** by the negative pressure vessel **204**.

(62) Referring now to FIG. **9**, a flow of gas during an embodiment of the calibration mode is depicted. As discussed herein, after the purging operation is performed, the positive pressure vessel **202** and the negative pressure vessel **204** are depressurized. Specifically, the pressure within the positive pressure vessel **202** and the negative pressure vessel **204** is depleted to 0 psi. As such, prior to performing another purging operation, the positive pressure vessel **202** and the negative pressure vessel **204** must be pressurized back to their calibrated state such that the positive pressure vessel **202** has a positive pressure and the negative pressure vessel **204** has a negative pressure, or is maintained at a vacuum. In embodiments, during the calibration mode, the gas within the negative pressure vessel **204** may be transferred to the positive pressure vessel **202** to create a vacuum within the negative pressure vessel **204** and positively pressurize the positive pressure vessel **202**. More particularly, as described herein, the dual piston assembly **244** may be operated to draw gas out of the negative pressure vessel **204** and direct the gas back into the positive pressure vessel **202**. In doing so, the second positive pressure valve **240** and the second negative pressure valve **242** are both positioned into the open state such that gas is permitted to flow therethrough. Additionally, the first piston valve **248** is positioned into the open state to allow air from the compressed air tank **246**

to be provided to and actuate the dual piston assembly **244**. The second piston valve **250** is also positioned into the open state to permit gas to flow into and out of the dual piston assembly **244**. The dual piston assembly **244** is operated until the positive pressure vessel **202** and the negative pressure vessel **204** are pressurized back into their respective calibrated state, which may be detected by the positive pressure sensor **230** and the negative pressure sensor **238**, respectively.

(63) Referring now to FIG. **10**, a flow of gas during another embodiment of the calibration mode is depicted. Rather than transferring gas from the negative pressure vessel **204** to the positive pressure vessel **202**, as discussed in the above embodiment of the calibration mode, external gas may be separately provided to the positive pressure vessel **202** and used to draw gas out of the negative pressure vessel **204**. Specifically, the second positive pressure valve **240** may be positioned into the open state and the external gas supply **252** may be operated to deliver gas through the external gas supply distributor **254** and into the positive pressure vessel **202**. It should be appreciated that, in the present embodiment, the external gas supply distributor **254** is configured to direct the flow of gas to only the positive pressure vessel **202** and not the negative pressure vessel **204**. Alternatively, the second negative pressure valve **242** may be configured to not receive gas from the external gas supply distributor **254**. In embodiments, the positive pressure sensor **230** may detect when the positive pressure vessel **202** is positioned back into the calibrated state and, in response, the second positive pressure valve **240** may be positioned into the closed state to prevent additional gas from entering the positive pressure vessel **202**. Additionally, the external gas supply **252** may be operated to direct a flow of gas to the Venturi device **256** when the first Venturi valve **258** is in the open state. Thereafter, the second negative pressure valve **242** and the second Venturi valve **260** are positioned into the open state such that the flow of gas through the Venturi device **256** draws gas out of the negative pressure vessel **204**. In embodiments, the negative pressure sensor **238** may detect when the negative pressure vessel **204** is positioned back into the calibrated state and, in response, the second negative pressure valve **242** may be positioned into the closed state to prevent additional gas from being drawn out of the negative pressure vessel **204**. Thereafter, the external gas supply **252** may be deactivated.

(64) Referring now to FIG. **11**, a flow of gas during the cleaning mode is depicted. In embodiments, it may be necessary to supply gas to both the positive pressure vessel **202** and the negative pressure vessel **204** to ensure that there are no leaks within the gas control system **106** or to inert the entire gas control system **106**. As such, the external gas supply **252** is operated to provide gas to both the positive pressure vessel **202** and the negative pressure vessel **204**. More particularly, the external gas supply distributor **254** is configured to direct the flow of gas to both the positive pressure vessel **202** and the negative pressure vessel **204**. Additionally, the second positive pressure valve **240** and the second negative pressure valve **242** are both positioned into the open position to allow the gas from the external gas supply **252** to enter the positive pressure vessel **202** and the negative pressure vessel **204**. After the cleaning mode is completed, the external gas supply **252** may be deactivated.

(65) From the above, it is to be appreciated that defined herein is a manufacturing apparatus including a printhead assembly including a first manifold, a second manifold, and one or more printheads, a binder control system, a gas control system, a power source, and a binder tank. The power source provides power to the binder control system, the gas control system, and the printhead assembly. The binder control system controls the flow of binder from the binder tank to each of the first manifold and the second manifold. The gas control system controls a flow of gas from an external gas supply to each of the first manifold and the second manifold. During a purging operation, gas may be rapidly circulated between the gas control system and the manifolds to purge a clogged nozzle of the one or more printheads.

(66) This provides a number of benefits such as, for example, improved safety, printhead stability, robustness, and controllability, and reduced costs and response time by rapidly pressurizing and purging. More particularly, the positive pressure vessel and the negative pressure vessel supply the

printheads with the required gas pressure necessary for back purging and vacuum-based meniscus control of the fluid. This also allows for ultra-precise printhead gas pressure control achieved by the first pressure control valve and the second pressure control valve. The piston assembly permits gas to flow to the positive pressure vessel within a fully closed gas circuit.

(67) Further aspects of the embodiments described herein are provided by the subject matter of the following clauses:

(68) Clause 1. A gas control system comprising: a positive pressure vessel; a negative pressure vessel; a first pressure control valve configured to control a first flow of gas to and from a first manifold of a printhead assembly; and a second pressure control valve configured to control a second flow of gas to and from a second manifold of the printhead assembly, wherein during a normal positive pressure mode, a primary flow of gas forming the first flow of gas and the second flow of gas flows from the positive pressure vessel to the first manifold and the second manifold through a respective one of the first pressure control valve and the second pressure control valve, wherein during a positive pressure purge mode, the first flow of gas and the second flow of gas from the positive pressure vessel bypasses the first pressure control valve and the second pressure control valve to flow to the first manifold and the second manifold.

(69) Clause 2. The gas control system of clause 1, wherein: during a negative pressure purge mode and a normal negative pressure mode, the first flow of gas and the second flow of gas is drawn from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve.

(70) Clause 3. The gas control system of clause 1 or clause 2, wherein: the positive pressure vessel is calibrated to a positive pressure greater than or equal to 10 psi and less than or equal to 20 psi; and the negative pressure vessel is calibrated to a negative pressure less than or equal to -10 psi and greater than or equal to -15 psi.

(71) Clause 4. The gas control system of any one of clauses 1-3, further comprising: a first positive pressure valve positionable between a first open state, a second open state, and a closed state; and a control side first manifold valve positionable between a first open state, a second open state, and a closed state to direct the primary flow of gas toward the first manifold, wherein when the first positive pressure valve is in the first open state, the primary flow of gas from the positive pressure vessel is directed to the control side first manifold valve, wherein when the first positive pressure valve is in the second open state, the primary flow of gas from the positive pressure vessel is directed to the first pressure control valve and the second pressure control valve.

(72) Clause 5. The gas control system of any one of clauses 1-4, further comprising: a first negative pressure valve positionable between an open state and a closed state, wherein when the first negative pressure valve is in the open state, a secondary flow of gas is permitted to be drawn into the negative pressure vessel from the first pressure control valve and the second pressure control valve.

(73) Clause 6. The gas control system of any one of clauses 1-5, further comprising: a control side second manifold valve positionable between a first open state, a second open state, and a closed state, wherein when the control side second manifold valve is in the first open state, the primary flow of gas from the positive pressure vessel is permitted to flow through the control side second manifold valve and to the second manifold, wherein when the control side second manifold valve is in the second open state, a secondary flow of gas is permitted to flow between the second manifold and the second pressure control valve.

(74) Clause 7. The gas control system of any one of clauses 1-6, further comprising: a positive pressure sensor configured to detect a pressure within the positive pressure vessel; a negative pressure sensor configured to detect a pressure within the negative pressure vessel; and an electronic control unit configured to receive data from the positive pressure sensor and the negative pressure sensor.

(75) Clause 8. The gas control system of any one of clauses 1-7, further comprising: a dual piston

assembly configured to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel.

(76) Clause 9. The gas control system of any one of clauses 1-8, further comprising: an external gas supply in direct fluid communication with the positive pressure vessel; and a Venturi device in fluid communication with the external gas supply, wherein, during a calibration mode, the external gas supply directs gas from the external gas supply into the positive pressure vessel to establish a positive pressure in the positive pressure vessel, wherein, during the calibration mode, the external gas supply directs gas from the external gas supply into the Venturi device, the Venturi device configured to draw gas out of the negative pressure vessel to establish a negative pressure in the negative pressure vessel.

(77) Clause 10. A manufacturing apparatus comprising: a printhead assembly including a first manifold, a second manifold, and a printhead, the first manifold and the second manifold directing a first flow of binder to and from the printhead, the first manifold and the second manifold each including a float for detecting a level of binder in a respective one of the first manifold and the second manifold; a binder control system for controlling a second flow of binder to the first manifold and the second manifold; and the gas control system of claim 1.

(78) Clause 11. The manufacturing apparatus of clause 10, wherein during a normal negative pressure mode and in response to the float detecting a level of binder falling below a predetermined threshold, the negative pressure vessel draws gas out of the first manifold and the second manifold and through a respective one of the first pressure control valve and the second pressure control valve.

(79) Clause 12. The manufacturing apparatus of clause 10 or clause 11, wherein the float of each of the first manifold and the second manifold comprises: an upper float configured to detect when the level of binder within a respective one of the first manifold and the second manifold rises above an upper predetermined threshold; and a lower float configured to detect when the level of binder within a respective one of the first manifold and the second manifold falls below above a lower predetermined threshold.

(80) Clause 13. The manufacturing apparatus of any one of clauses 10-12, wherein: during a negative pressure purge mode and a normal negative pressure mode, gas is drawn from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve.

(81) Clause 14. The manufacturing apparatus of any one of clauses 10-13, wherein: the positive pressure vessel is calibrated to a positive pressure to deliver gas to the first manifold and the second manifold; and the negative pressure vessel is calibrated to a negative pressure to draw gas out of the first manifold and the second manifold.

(82) Clause 15. The manufacturing apparatus of any one of clauses 10-14, further comprising: a dual piston assembly configured to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel during a calibration mode.

(83) Clause 16. The manufacturing apparatus of any one of clauses 10-15, further comprising: an external gas supply in direct fluid communication with the positive pressure vessel; and a Venturi device in fluid communication with the external gas supply, wherein, during a calibration mode, the external gas supply directs gas into the positive pressure vessel to establish a positive pressure in the positive pressure vessel, wherein, during the calibration mode, the external gas supply directs gas into the Venturi device, the Venturi device configured to draw gas out of the negative pressure vessel to establish a negative pressure in the negative pressure vessel.

(84) Clause 17. A method comprising: pressurizing a positive pressure vessel of a gas control system to a positive pressure; pressurizing a negative pressure vessel of the gas control system to a negative pressure to create a vacuum within the negative pressure vessel; moving gas between the

positive pressure vessel and the negative pressure vessel to achieve a fully closed gas circuit; operating the gas control system to perform a normal operation to maintain a level of binder within a first manifold and a second manifold, the normal operation includes a normal positive pressure normal mode in which gas from the positive pressure vessel is directed to a first manifold through a first pressure control valve, and gas from the positive pressure vessel is directed to a second manifold through a second pressure control valve; and operating the gas control system to perform a purging operation to de-clog a nozzle of a printhead, the purging operation including a positive pressure purge mode in which gas from the positive pressure vessel is permitted to flow to the first manifold and the second manifold while bypassing the first pressure control valve and the second pressure control valve.

(85) Clause 18. The method of clause 17, further comprising: drawing gas from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve during a negative pressure purge mode of the purging operation and a normal negative pressure mode of the normal operation, wherein the gas is an inert gas and the binder in the first manifold and the second manifold is a solvent-based binder.

(86) Clause 19. The method of clause 17 or clause 18, further comprising: calibrating the positive pressure vessel to a positive pressure prior to the purging operation and the normal operation to permit gas to be delivered to the first manifold and the second manifold; and calibrating the negative pressure vessel to a negative pressure prior to the purging operation and the normal operation to permit gas to be drawn out of the first manifold and the second manifold.

(87) Clause 20. The method of any one of clauses 17-19, further comprising: actuating a dual piston assembly to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel during a calibration mode. While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A gas control system comprising: a positive pressure vessel; a negative pressure vessel; a first pressure control valve configured to control a first flow of gas to and from a first manifold of a printhead assembly; and a second pressure control valve configured to control a second flow of gas to and from a second manifold of the printhead assembly, wherein during a normal positive pressure mode, a primary flow of gas forming the first flow of gas and the second flow of gas flows from the positive pressure vessel to the first manifold and the second manifold through a respective one of the first pressure control valve and the second pressure control valve, wherein during a positive pressure purge mode, the first flow of gas and the second flow of gas from the positive pressure vessel bypasses the first pressure control valve and the second pressure control valve to flow to the first manifold and the second manifold.

2. The gas control system of claim 1, wherein: during a negative pressure purge mode and a normal negative pressure mode, the first flow of gas and the second flow of gas is drawn from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve.

3. The gas control system of claim 1, wherein: the positive pressure vessel is calibrated to a positive pressure greater than or equal to 10 psi and less than or equal to 20 psi; and the negative

pressure vessel is calibrated to a negative pressure less than or equal to -10 psi and greater than or equal to -15 psi.

4. The gas control system of claim 1, further comprising: a first positive pressure valve positionable between a first open state, a second open state, and a closed state; and a control side first manifold valve positionable between a first open state, a second open state, and a closed state to direct the primary flow of gas toward the first manifold, wherein when the first positive pressure valve is in the first open state, the primary flow of gas from the positive pressure vessel is directed to the control side first manifold valve, wherein when the first positive pressure valve is in the second open state, the primary flow of gas from the positive pressure vessel is directed to the first pressure control valve and the second pressure control valve.

5. The gas control system of claim 1, further comprising: a first negative pressure valve positionable between an open state and a closed state, wherein when the first negative pressure valve is in the open state, a secondary flow of gas is permitted to be drawn into the negative pressure vessel from the first pressure control valve and the second pressure control valve.

6. The gas control system of claim 1, further comprising: a control side second manifold valve positionable between a first open state, a second open state, and a closed state, wherein when the control side second manifold valve is in the first open state, the primary flow of gas from the positive pressure vessel is permitted to flow through the control side second manifold valve and to the second manifold, wherein when the control side second manifold valve is in the second open state, a secondary flow of gas is permitted to flow between the second manifold and the second pressure control valve.

7. The gas control system of claim 1, further comprising: a positive pressure sensor configured to detect a pressure within the positive pressure vessel; a negative pressure sensor configured to detect a pressure within the negative pressure vessel; and an electronic control unit configured to receive data from the positive pressure sensor and the negative pressure sensor.

8. The gas control system of claim 1, further comprising: a dual piston assembly configured to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel.

9. The gas control system of claim 1, further comprising: an external gas supply in direct fluid communication with the positive pressure vessel; and a Venturi device in fluid communication with the external gas supply, wherein, during a calibration mode, the external gas supply directs gas from the external gas supply into the positive pressure vessel to establish a positive pressure in the positive pressure vessel, wherein, during the calibration mode, the external gas supply directs gas from the external gas supply into the Venturi device, the Venturi device configured to draw gas out of the negative pressure vessel to establish a negative pressure in the negative pressure vessel.

10. A manufacturing apparatus, comprising: a printhead assembly including a first manifold, a second manifold, and a printhead, the first manifold and the second manifold directing a first flow of binder to and from the printhead, the first manifold and the second manifold each including a float for detecting a level of binder in a respective one of the first manifold and the second manifold; and the gas control system of claim 1.

11. The manufacturing apparatus of claim 10, wherein during a normal negative pressure mode and in response to the float detecting a level of binder falling below a predetermined threshold, the negative pressure vessel draws gas out of the first manifold and the second manifold and through a respective one of the first pressure control valve and the second pressure control valve.

12. The manufacturing apparatus of claim 10, wherein the float of each of the first manifold and the second manifold comprises: an upper float configured to detect when the level of binder within a respective one of the first manifold and the second manifold rises above an upper predetermined threshold; and a lower float configured to detect when the level of binder within a respective one of the first manifold and the second manifold falls below a lower predetermined threshold.

13. The manufacturing apparatus of claim 10, wherein: during a negative pressure purge mode and a normal negative pressure mode, gas is drawn from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve.

14. The manufacturing apparatus of claim 10, wherein: the positive pressure vessel is calibrated to a positive pressure greater than or equal to 10 psi and less than or equal to 20 psi; and the negative pressure vessel is calibrated to a negative pressure less than or equal to -10 psi and greater than or equal to -15 psi.

15. The manufacturing apparatus of claim 10, further comprising: a dual piston assembly configured to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel during a calibration mode.

16. The manufacturing apparatus of claim 10, further comprising: an external gas supply in direct fluid communication with the positive pressure vessel; and a Venturi device in fluid communication with the external gas supply, wherein, during a calibration mode, the external gas supply directs gas into the positive pressure vessel to establish a positive pressure in the positive pressure vessel, wherein, during the calibration mode, the external gas supply directs gas into the Venturi device, the Venturi device configured to draw gas out of the negative pressure vessel to establish a negative pressure in the negative pressure vessel.

17. A method comprising: pressurizing a positive pressure vessel of a gas control system to a positive pressure; pressurizing a negative pressure vessel of the gas control system to a negative pressure to create a vacuum within the negative pressure vessel; moving gas between the positive pressure vessel and the negative pressure vessel to achieve a fully closed gas circuit; operating the gas control system to perform a normal operation to maintain a level of binder within a first manifold and a second manifold, the normal operation includes a normal positive pressure normal mode in which gas from the positive pressure vessel is directed to a first manifold through a first pressure control valve, and gas from the positive pressure vessel is directed to a second manifold through a second pressure control valve; and operating the gas control system to perform a purging operation to de-clog a nozzle of a printhead, the purging operation including a positive pressure purge mode in which gas from the positive pressure vessel is permitted to flow to the first manifold and the second manifold while bypassing the first pressure control valve and the second pressure control valve.

18. The method of claim 17, further comprising: drawing gas from the first manifold and the second manifold to the negative pressure vessel through a respective one of the first pressure control valve and the second pressure control valve during a negative pressure purge mode of the purging operation and a normal negative pressure mode of the normal operation, wherein the gas is an inert gas and the binder in the first manifold and the second manifold is a solvent-based binder.

19. The method of claim 17, further comprising: calibrating the positive pressure vessel to a positive pressure prior to the purging operation and the normal operation to permit gas to be delivered to the first manifold and the second manifold; and calibrating the negative pressure vessel to a negative pressure prior to the purging operation and the normal operation to permit gas to be drawn out of the first manifold and the second manifold.

20. The method of claim 17, further comprising: actuating a dual piston assembly to draw gas out of the negative pressure vessel and direct the gas back into the positive pressure vessel to establish a negative pressure in the negative pressure vessel and a positive pressure in the positive pressure vessel during a calibration mode.
